

Enroll No.....

Class Roll No.....

Second Mid Semester Examination

Data Science Toolkit (CS-601)

Branch: CSE Semester: VI

Time 1:30 Hrs

Max. Marks 20

Q.1

- a) List any four important components of the **PyTorch** framework used for deep learning projects.
- b) How does **text generation in PyTorch** differ from sentiment analysis in terms of model objectives?
(CO3 – 7 marks)

OR

- a) A student is using **Keras** to perform **style transfer** between two images. Identify the high-level steps and which model architecture is typically used.
- b) Analyze the challenges in **time series forecasting** using **ARIMA** when dealing with non-stationary data.

Q.2

- a) Name any two **data structures in R** that are commonly used for data analysis.
- b) Explain the process of **importing data into R** and visualizing it using basic plots.
(CO4 – 7 marks)

OR

- a) How would you perform **classification in R** using a sample dataset? Mention the steps and packages used.
- b) Compare **matrices and data frames** in R. When would you use one over the other?

Q.3

- a) Define **Power BI** and name two important features that support business intelligence dashboards.
- b) How does **Google Colab** differ from local IDEs in terms of data science project development?
(CO5 – 6 marks)

OR

- a) You are assigned a case study project using **Weka** and **RapidMiner**. Briefly outline two tasks you would perform with each tool.
- b) Analyze the benefits of using **version control tools** like **GitHub** in collaborative data science projects.